Mathematics II

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11/03/2026

You can find the last version of these course materials at

https://mbujosab.github.io/MatematicasII/

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1 Highlights of **Lesson 11**

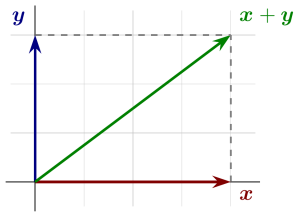
Highlights of Lesson 11

- Orthogonal vectors and subspaces
- Null space $\perp$ row space
$$\mathcal{N}\left(\mathbf{A}\right) \perp \mathcal{C}\left(\mathbf{A}^{\intercal}\right)$$
- Left null space $\perp$ column space
$$\mathcal{N}\left(\mathbf{A}^{\intercal}\right) \perp \mathcal{C}\left(\mathbf{A}\right)$$

2 Some definitions

- Dot product
$$\boldsymbol{a} \cdot \boldsymbol{b} \;=\; \sum_{i=1}^{n} a_i b_i$$
- Length of a vector $\|\boldsymbol{a}\| = \sqrt{\boldsymbol{a} \cdot \boldsymbol{a}}$ $\qquad \boldsymbol{a} \cdot \boldsymbol{a} = \|\boldsymbol{a}\|^2.$
- Unit vector: $\|\boldsymbol{a}\| = 1$ $\qquad \frac{1}{\|\boldsymbol{x}\|} \cdot \boldsymbol{x}$
- Orthogonal (perpendicular) vectors: $\boldsymbol{x} \cdot \boldsymbol{y} = 0.$

3 Orthogonal vectors



$$\boldsymbol{x}\cdot\boldsymbol{y} = 0 \iff \boldsymbol{x}\perp\boldsymbol{y}$$

Pythagorean Thm.: $\boldsymbol{x}\cdot\boldsymbol{y} = 0 \iff \|\boldsymbol{x}\|^2+\|\boldsymbol{y}\|^2 = \|\boldsymbol{x}+\boldsymbol{y}\|^2$

$$\boldsymbol{x}\cdot\boldsymbol{x}+\boldsymbol{y}\cdot\boldsymbol{y} = (\boldsymbol{x}+\boldsymbol{y})\cdot(\boldsymbol{x}+\boldsymbol{y}).$$

4 Squared length of a vector

$$\|\boldsymbol{v}\|^2 = \boldsymbol{v}\cdot\boldsymbol{v}$$

$$\boldsymbol{x} = \begin{pmatrix}1\\2\\3\end{pmatrix} \quad\to\quad \|\boldsymbol{x}\|^2 = \qquad ; \qquad \boldsymbol{y} = \begin{pmatrix}2\\-1\\0\end{pmatrix} \quad\to\quad \|\boldsymbol{y}\|^2 = \qquad ;$$

Are these vectors orthogonal?

$$\boldsymbol{x}+\boldsymbol{y} = \begin{pmatrix}\\ \\ \end{pmatrix}; \qquad \|\boldsymbol{x}+\boldsymbol{y}\|^2 = \qquad ;$$

(Pythagorean T.) (Orthogonality)

$$\boldsymbol{x}\cdot\boldsymbol{x}+\boldsymbol{y}\cdot\boldsymbol{y} = (\boldsymbol{x}+\boldsymbol{y})\cdot(\boldsymbol{x}+\boldsymbol{y}) \iff \boldsymbol{x}\cdot\boldsymbol{y} = 0.$$

5 Orthogonal subspaces

When the subspace $\mathcal{S}$ is orthogonal to subspace $\mathcal{T}$:

$\boxed{\text{Every vector in } \mathcal{S} \text{ is orthogonal to every vector in } \mathcal{T}}$

Are the plane of the *blackboard* and the floor orthogonal?

6 Null space orthogonal to row space

- $\mathcal{N}(\mathbf{A}) \perp$ rows of $\mathbf{A}$

$$\mathbf{A}\boldsymbol{x} = \mathbf{0} \implies \begin{pmatrix}({}_{1|}\mathbf{A})\cdot\boldsymbol{x}\\ \vdots \\ ({}_{m|}\mathbf{A})\cdot\boldsymbol{x}\end{pmatrix} = \begin{pmatrix}0\\ \vdots \\ 0\end{pmatrix}$$

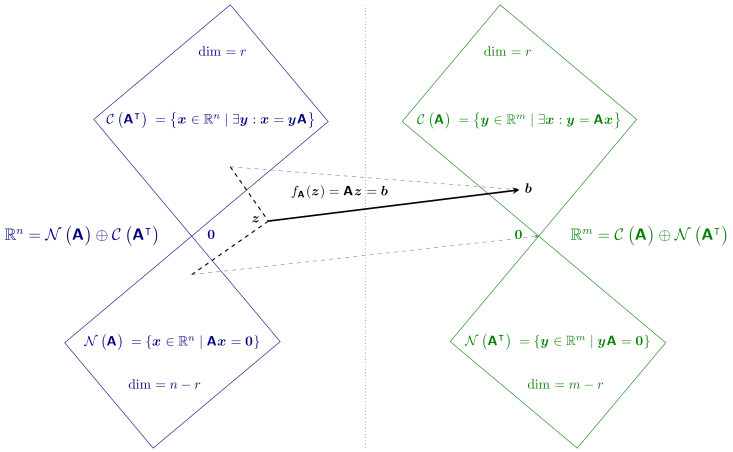
- $\mathcal{N}(\mathbf{A}) \perp \boldsymbol{d}\mathbf{A}, \quad \forall \boldsymbol{d}\in\mathbb{R}^m$ (any linear combination of the rows)

$$\boldsymbol{x}\in\mathcal{N}(\mathbf{A}) \quad\Rightarrow\quad \boldsymbol{d}\mathbf{A}\boldsymbol{x} = \boldsymbol{d}\cdot\mathbf{0} = 0.$$

$\boxed{\text{null space} \perp \text{row space} \qquad \mathcal{N}(\mathbf{A}) \perp \mathcal{C}(\mathbf{A}^\intercal)}$

Also: $\boldsymbol{x}\mathbf{A} = \mathbf{0} \quad\Rightarrow\quad \mathcal{N}(\mathbf{A}^\intercal) \perp \mathcal{C}(\mathbf{A})$

7 The big picture: direct sum of orthogonal complements



$$\mathcal{C}(\mathbf{A}^\intercal) \perp \mathcal{N}(\mathbf{A}) \qquad \mathcal{C}(\mathbf{A}) \perp \mathcal{N}(\mathbf{A}^\intercal)$$

$$\boldsymbol{f} \cdot \boldsymbol{x} = \boldsymbol{y}\mathbf{A}\boldsymbol{x} = \boldsymbol{y} \cdot \mathbf{0} \qquad \boldsymbol{y} \cdot \boldsymbol{b} = \boldsymbol{y}\mathbf{A}\boldsymbol{x} = \mathbf{0} \cdot \boldsymbol{x}$$

8 Revisiting Gaussian elimination

An algorithm to find a basis for the orthogonal complement

Give me some vectors (I write them as rows of **M**) and ...

$$\begin{bmatrix}\mathbf{M}\\ \hline \mathbf{I}\end{bmatrix} = \begin{bmatrix}1 & -3 & 0 & -1\\ 0 & -1 & 1 & 1\\ 1 & -4 & 1 & 0\\ \hline 1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 1\end{bmatrix} \xrightarrow[\substack{[(3)\mathbf{1}+\mathbf{2}]\\ [(1)\mathbf{1}+\mathbf{4}]\\ [(1)\mathbf{2}+\mathbf{3}]\\ [(1)\mathbf{2}+\mathbf{4}]}]{\tau} \begin{bmatrix}1 & 0 & 0 & 0\\ 0 & -1 & 0 & 0\\ 1 & -1 & 0 & 0\\ \hline 1 & 3 & 3 & 4\\ 0 & 1 & 1 & 1\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 1\end{bmatrix} = \begin{bmatrix}\mathbf{L}\\ \hline \mathbf{E}\end{bmatrix} = \begin{bmatrix}\mathbf{C} & \mathbf{0}\\ \hline \mathbf{D} & \mathbf{N}\end{bmatrix}$$

Basis for the span of the given (row) vectors: $\mathcal{V}$

Basis for the orthogonal complement: $\mathcal{V}^\perp$ $\qquad \mathbf{M}\mathbf{N} = \mathbf{0}$

But if you had given me $\mathbf{N}_{|1}$ and $\mathbf{N}_{|2}$, and I start anew... I would obtain a basis for...

Questions of the Lecture 11

(L-11) Question 1. Describe the set of vectors in $\mathbb{R}^3$ orthogonal to $\begin{pmatrix}1\\3\\-1\end{pmatrix}$

(Hefferon, 2008, exercise 2.15 from section II.2.)

(L-11) Question 2. Is there any vector that is perpendicular to itself?

(L-11) Question 3. Find the length of each vector

(a) $\begin{pmatrix}1\\3\end{pmatrix}$. (b) $\begin{pmatrix}-1\\2\end{pmatrix}$. (c) $\begin{pmatrix}4\\1\\1\end{pmatrix}$.

(d) $\begin{pmatrix}0\\0\\0\end{pmatrix}$. (e) $\begin{pmatrix}1\\-1\\1\\0\end{pmatrix}$.

(Hefferon, 2008, exercise 2.11 from section II.2.)

(L-11) Question 4. Find a unit vector with the same direction as $\boldsymbol{v} = (2, -1, 0, 4, -2)$.

(L-11) Question 5. Find k so that these two vectors are perpendicular.

$$(k, \quad 1,), \qquad (4, \quad 3,).$$

(Hefferon, 2008, exercise 2.14 from section II.2.)

(L-11) Question 6. Construct a matrix with the required property or state why that is impossible:

(a) Column space contains $\begin{pmatrix}1\\2\\-3\end{pmatrix}$ and $\begin{pmatrix}2\\-3\\5\end{pmatrix}$, the null space contains $\begin{pmatrix}1\\1\\1\end{pmatrix}$

(b) Row space contains $\begin{pmatrix}1\\2\\-3\end{pmatrix}$ and $\begin{pmatrix}2\\-3\\5\end{pmatrix}$, and the null space contains $\begin{pmatrix}1\\1\\1\end{pmatrix}$

(c) $\mathbf{A}\boldsymbol{x} = \begin{pmatrix}1\\1\\1\end{pmatrix}$ has a solution and $\mathbf{A}^\intercal \begin{pmatrix}1\\0\\0\end{pmatrix} = \begin{pmatrix}0\\0\\0\end{pmatrix}$

(d) Each row is orthogonal to every column (and **A** is not the zero matrix)

(e) Columns add up to a column of zeros; rows add up to a row of 1's.

(Strang, 2003, exercise 3 from section 4.1.)

(L-11) Question 7. If $\mathbf{AB} = \mathbf{0}$, the columns of **B** are in the _____ of **A**. The rows of **A** are in the _____ of **B**. Why can't **A** and **B** be 3 by 3 matrices of rank 2?

(Strang, 2003, exercise 4 from section 4.1.)

(L-11) Question 8. Suppose that $\boldsymbol{u} \cdot \boldsymbol{v} = \boldsymbol{u} \cdot \boldsymbol{w}$ and $\boldsymbol{u} \neq \mathbf{0}$. Must $\boldsymbol{v} = \boldsymbol{w}$?

(Hefferon, 2008, exercise 2.20 from section II.2.)

(L-11) Question 9.

(a) If $\mathbf{A}\boldsymbol{x} = \boldsymbol{b}$ has a solution and $\mathbf{A}^\intercal \boldsymbol{y} = \mathbf{0}$, then $\boldsymbol{y}$ is perpendicular to ____.
(b) If $\mathbf{A}^\intercal \boldsymbol{y} = \boldsymbol{c}$ has a solution and $\mathbf{A}\boldsymbol{x} = \mathbf{0}$, then $\boldsymbol{x}$ is perpendicular to ____.

(Strang, 2003, exercise 5 from section 4.1.)

(L-11) Question 10. Demonstrate, for $\mathbb{R}^n$, that if $\boldsymbol{u}$ and $\boldsymbol{v}$ are perpendicular then $||\boldsymbol{u}+\boldsymbol{v}||^2 = ||\boldsymbol{u}||^2 + ||\boldsymbol{v}||^2$.
(Hefferon, 2008, exercise 2.33 from section II.2.)

(L-11) Question 11.

(a) Find a 1 by 3 matrix whose null space consists of all vectors in $\mathbb{R}^3$ such that $x_1 + 2x_2 + 4x_3 = 0$. Find a 3 by 3 matrix with that same null space.
(b) Find a 1 by 3 matrix whose null space is orthogonal to all vectors in $\mathbb{R}^3$ such that $x_1 + 2x_2 + 4x_3 = 0$. Find a 3 by 3 matrix with that same null space.

(Strang, 2006, exercise 9 from section 2.4.)

(L-11) Question 12. Consider $\underset{4\times 2}{\mathbf{A}}$ with exactly two special solutions for $\boldsymbol{x}\mathbf{A} = \mathbf{0}$:

$\boldsymbol{s}_1 = \begin{pmatrix}3, & 1, & 0, & 0,\end{pmatrix}$, and $\boldsymbol{s}_2 = \begin{pmatrix}6, & 0, & 2, & 1,\end{pmatrix}$.

(a) Find the reduced row echelon form $\mathbf{R}$ of $\mathbf{A}$.
(b) What is the row space of $\mathbf{A}$?
(c) What is the complete solution to $\boldsymbol{x}\mathbf{R} = \begin{pmatrix}3, & 6,\end{pmatrix}$?

(d) Find a combination of rows 2, 3, 4 that equals $\mathbf{0}$. (Not OK to use $0({}_{2|}\mathbf{A}) + 0({}_{3|}\mathbf{A}) + 0({}_{4|}\mathbf{A})$. The problem is to show that these rows are dependent.)

(L-11) Question 13. Suppose $\mathbf{A}\boldsymbol{x} = \boldsymbol{b}$ has a solution (maybe many solutions). It can be shown that any solution $\boldsymbol{x}$ of this system can be decomposed as the sum of two vectors ($\boldsymbol{x} = \boldsymbol{x}_r + \boldsymbol{x}_n$) where $\boldsymbol{x}_r$ is a combination of the rows of $\mathbf{A}$ and $\boldsymbol{x}_n$ belongs to the solution set of $\mathbf{A}\boldsymbol{x} = \mathbf{0}$.

(a) (0.5^{pts}) Prove that $\mathbf{A}(\boldsymbol{x}_r) = \boldsymbol{b}$.
(b) (1^{pts}) Suppose that $\boldsymbol{v}_r$ is a linear combination of the rows of $\mathbf{A}$ and furthermore $\mathbf{A}(\boldsymbol{v}_r) = \boldsymbol{b}$. What vector subspaces does the difference $(\boldsymbol{v}_r - \boldsymbol{x}_r)$ belong to? Show that $\boldsymbol{x}_r$ and $\boldsymbol{v}_r$ are equal.
(c) (1^{pts}) Compute the solution $\boldsymbol{x}_r$ in the row space of this matrix $\mathbf{A}$, by solving for c and d

$$\begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & -1 \end{bmatrix} \boldsymbol{x}_r = \begin{pmatrix} 14 \\ 9 \end{pmatrix} \quad \text{with} \quad \boldsymbol{x}_r = c\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + d\begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}.$$

1 Highlights of **Lecture 12**

Highlights of Lecture 12

- From parametric to Cartesian (or implicit) equations
- Choosing among parametric equations

2 Cartesian (implicit) and parametric equations of lines and planes

Cartesian (implicit) equations $\{\boldsymbol{x} \in \mathbb{R}^n \mid \mathbf{A}\boldsymbol{x} = \boldsymbol{b}\}$:

For example

$$\left\{ \boldsymbol{x} \in \mathbb{R}^3 \;\middle|\; \begin{bmatrix} 1 & -1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \boldsymbol{x} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\} = \text{sol. set of } \begin{cases} x_1 - x_2 + x_3 = 1 \\ x_3 = 1 \end{cases}$$

Parametric equations:

for the above set

$$\left\{ \boldsymbol{x} \in \mathbb{R}^3 \;\middle|\; \exists \boldsymbol{p} \in \mathbb{R}^1 : \boldsymbol{x} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \boldsymbol{p} \right\}$$

In this case *dimension* 1 line

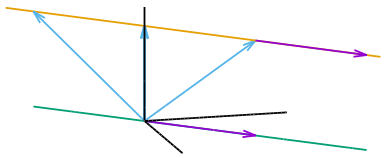
A line (there is only one parameter a) line

or

$$\left\{ \boldsymbol{x} \in \mathbb{R}^3 \;\middle|\; \exists \boldsymbol{p} \in \mathbb{R}^1 : \boldsymbol{x} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \boldsymbol{p} \right\}$$

or

$$\left\{ \boldsymbol{x} \in \mathbb{R}^3 \;\middle|\; \exists \boldsymbol{p} \in \mathbb{R}^1 : \boldsymbol{x} = \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \boldsymbol{p} \right\}$$



3 Cartesian (implicit) and parametric equations of lines and planes

Cartesian (implicit) equations $\{\boldsymbol{x} \in \mathbb{R}^n \mid \mathbf{A}\boldsymbol{x} = \boldsymbol{b}\}$:

For example

$$\{\boldsymbol{x} \in \mathbb{R}^3 \mid \begin{bmatrix} 1 & -1 & 1 \end{bmatrix} \boldsymbol{x} = (1,)\} = \text{sol. set of } \{x_1 - x_2 + x_3 = 1$$

Parametric equations:

for the above set

$$\left\{ \boldsymbol{x} \in \mathbb{R}^3 \;\middle|\; \exists \boldsymbol{p} \in \mathbb{R}^2 : \boldsymbol{x} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} + \begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \boldsymbol{p} \right\}$$

In this case *dimension* 2 A plane (there are two parameters a and b)

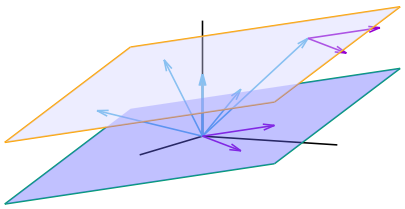
plane plane

or

$$\left\{ \boldsymbol{x} \in \mathbb{R}^3 \;\middle|\; \exists \boldsymbol{p} \in \mathbb{R}^2 : \boldsymbol{x} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \boldsymbol{p} \right\}$$

but also

$$\left\{ \boldsymbol{x} \in \mathbb{R}^3 \;\middle|\; \exists \boldsymbol{p} \in \mathbb{R}^2 : \boldsymbol{x} = \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix} + \begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \boldsymbol{p} \right\}$$



4 From parametric to Cartesian equations

$$\boxed{\mathcal{C}\left(\mathbf{A}^\intercal\right) \perp \mathcal{N}\left(\mathbf{A}\right)}$$

Consider

$$H = \left\{ \boldsymbol{x} \in \mathbb{R}^n \;\middle|\; \exists \boldsymbol{p} \in \mathbb{R}^k : \boldsymbol{x} = \boldsymbol{s} + \begin{bmatrix} \boldsymbol{n}_1; & \dots & \boldsymbol{n}_k; \end{bmatrix} \boldsymbol{p} \right\}.$$

If we find $\mathbf{A}$ such that $\mathbf{A}\boldsymbol{n}_i = \mathbf{0}$, then if $\boldsymbol{x} \in H$

$$\mathbf{A}\boldsymbol{x} = \mathbf{A}\boldsymbol{s} + \underbrace{\mathbf{A}\begin{bmatrix} \boldsymbol{n}_1; & \dots & \boldsymbol{n}_k; \end{bmatrix}}_{\mathbf{0}} \boldsymbol{p} \quad \Rightarrow \quad \mathbf{A}\boldsymbol{x} = \boldsymbol{b}, \quad \text{where } \boldsymbol{b} = \mathbf{A}\boldsymbol{s}.$$

Therefore

$$H = \{\boldsymbol{x} \in \mathbb{R}^n \mid \mathbf{A}\boldsymbol{x} = \boldsymbol{b}\}.$$

5 From the set of solutions to a linear system

Find the implicit equations of the plane P parallel to the span of $(1,2,0,-2)$ and $(0,0,1,3)$, that goes through $\boldsymbol{s}=(1,3,1,1)$.

$$P=\left\{\begin{pmatrix}x\\y\\z\\w\end{pmatrix}\;\middle|\;\exists a,b\in\mathbb{R}:\begin{pmatrix}x\\y\\z\\w\end{pmatrix}=\begin{pmatrix}1\\3\\1\\1\end{pmatrix}+a\begin{pmatrix}1\\2\\0\\-2\end{pmatrix}+b\begin{pmatrix}0\\0\\1\\3\end{pmatrix}\right\}$$

$$=\left\{\boldsymbol{x}\in\mathbb{R}^4\;\middle|\;\exists\boldsymbol{p}\in\mathbb{R}^2:\boldsymbol{x}=\begin{pmatrix}1\\3\\1\\1\end{pmatrix}+\begin{bmatrix}1&0\\2&0\\0&1\\-2&3\end{bmatrix}\boldsymbol{p}\right\}$$

We need vectors perpendicular to $(1,2,0,-2)$ and $(0,0,1,3)$

6 From the set of solutions to a linear system

$$\boldsymbol{x}=(x,y,z,w,);\qquad \boldsymbol{s}=(1,3,1,1,).$$

$$\left[\begin{array}{cccc}1&2&0&-2\\0&0&1&3\\\hline x&y&z&w\\\hline 1&3&1&1\end{array}\right]\xrightarrow[\substack{[(-2)\mathbf{1}+\mathbf{2}]\\[(2)\mathbf{1}+\mathbf{4}]}]{\boldsymbol{\tau}}\left[\begin{array}{cccc}1&0&0&0\\0&0&1&3\\\hline x&y-2x&z&w+2x\\\hline 1&1&1&3\end{array}\right]\xrightarrow[[(-3)\mathbf{3}+\mathbf{4}]]{\boldsymbol{\tau}}\left[\begin{array}{cccc}1&0&0&0\\0&0&1&0\\\hline x&y-2x&z&w+2x-3z\\\hline 1&1&1&0\end{array}\right]$$

So $\mathbf{A}=\begin{bmatrix}-2&1&0&0\\2&0&-3&1\end{bmatrix}$; and then $\mathbf{A}\boldsymbol{x}=\begin{pmatrix}-2x+y\\2x+w-3z\end{pmatrix}$ and $\boldsymbol{b}=\mathbf{A}\boldsymbol{s}=\begin{pmatrix}1\\0\end{pmatrix}$. Hence $\begin{cases}-2x+y=1\\2x-3z+w=0\end{cases}$

$$P=\left\{\boldsymbol{x}\in\mathbb{R}^4\;\middle|\;\begin{bmatrix}-2&1&0&0\\2&0&-3&1\end{bmatrix}\boldsymbol{x}=\begin{pmatrix}1\\0\end{pmatrix}\right\}.$$

7 A problem from Microeconomics

Solve for Y in terms of X to obtain the PPF

$$\begin{cases}X=4L_x\\Y=3L_y\\L_x+L_y=80\end{cases}\rightarrow\begin{cases}X-4L_x=0\\Y-3L_y=0\\L_x+L_y=80\end{cases}$$

("in terms of" X means X free)

$$\left[\begin{array}{cccc|c}1&0&-4&0&0\\0&1&0&-3&0\\0&0&1&1&-80\\\hline 1&0&0&0&0\\0&1&0&0&0\\0&0&1&0&0\\0&0&0&1&0\\\hline 0&0&0&0&1\end{array}\right]\xrightarrow[\substack{[(4)\mathbf{1}+\mathbf{3}]\\[(3)\mathbf{2}+\mathbf{4}]}]{\boldsymbol{\tau}}\left[\begin{array}{cccc|c}1&0&0&0&0\\0&1&0&0&0\\0&0&1&1&-80\\\hline 1&0&4&0&0\\0&1&0&3&0\\0&0&1&0&0\\0&0&0&1&0\\\hline 0&0&0&0&1\end{array}\right]\xrightarrow[\substack{[(-1)\mathbf{3}+\mathbf{4}]\\[(80)\mathbf{3}+\mathbf{5}]}]{\boldsymbol{\tau}}\left[\begin{array}{cccc|c}1&0&0&0&0\\0&1&0&0&0\\0&0&1&0&0\\\hline 1&0&4&-4&320\\0&1&0&3&0\\0&0&1&-1&80\\0&0&0&1&0\\\hline 0&0&0&0&1\end{array}\right]$$

$$\begin{pmatrix}X\\Y\\L_x\\L_y\end{pmatrix}=\begin{pmatrix}320\\0\\80\\0\end{pmatrix}+a\begin{pmatrix}-4\\3\\-1\\1\end{pmatrix}\Rightarrow a=L_y\Rightarrow\begin{pmatrix}X\\Y\\L_x\\L_y\end{pmatrix}=\begin{pmatrix}320-4L_y\\3L_y\\80-L_y\\L_y\end{pmatrix}\quad\text{"in terms of" } L_y$$

8 Free variable

$$\left[\begin{array}{cccc|c}1&0&0&0&0\\0&1&0&0&0\\0&0&1&0&0\\\hline 1&0&4&-4&320\\0&1&0&3&0\\0&0&1&-80&80\\0&0&0&1&0\\\hline 0&0&0&0&1\end{array}\right]\xrightarrow[\substack{\left[\left(\frac{-1}{4}\right)\mathbf{4}\right]\\[(-320)\mathbf{4}+\mathbf{5}]}]{\boldsymbol{\tau}}\left[\begin{array}{cccc|c}1&0&0&0&0\\0&1&0&0&0\\0&0&1&0&0\\\hline 1&0&4&1&0\\0&1&0&-3/4&240\\0&0&1&1/4&0\\0&0&0&-1/4&80\\\hline 0&0&0&0&1\end{array}\right]$$

$$\begin{pmatrix}X\\Y\\L_x\\L_y\end{pmatrix}=\begin{pmatrix}0\\240\\0\\80\end{pmatrix}+a\begin{pmatrix}1\\-\frac{3}{4}\\\frac{1}{4}\\-\frac{1}{4}\end{pmatrix}\Rightarrow a=X\Rightarrow\begin{pmatrix}X\\Y\\L_x\\L_y\end{pmatrix}=\begin{pmatrix}X\\240-\frac{3}{4}X\\\frac{1}{4}X\\80-\frac{1}{4}X\end{pmatrix}$$

"in terms of" X

9 Free variables

$$\begin{cases} x + 2y - z + w = -1 \\ -x - 2y + 3z + 5w = -5 \\ -x - 2y - z - 7w = 7 \end{cases}$$

1. Solve in terms of y and w
2. Solve in terms of x and w
3. Solve in terms of x and z
4. Solve in terms of x and y

$$\left[\begin{array}{cccc|c} 1 & 2 & -1 & 1 & -1 \\ -1 & -2 & 3 & 5 & -5 \\ -1 & -2 & -1 & -7 & 7 \\ \hline 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ \hline 0 & 0 & 0 & 0 & 1 \end{array}\right] \xrightarrow[\substack{[(-2)\mathbf{1}+\mathbf{2}] \\ [(1)\mathbf{1}+\mathbf{3}] \\ [(-1)\mathbf{1}+\mathbf{4}] \\ [(1)\mathbf{1}+\mathbf{5}]}]{\boldsymbol{\tau}} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 2 & 6 & -6 \\ -1 & 0 & -2 & -6 & 6 \\ \hline 1 & -2 & 1 & -1 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ \hline 0 & 0 & 0 & 0 & 1 \end{array}\right]$$

$$\xrightarrow[\substack{[(-3)\mathbf{3}+\mathbf{4}] \\ [(3)\mathbf{3}+\mathbf{5}]}]{\boldsymbol{\tau}} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 2 & 0 & 0 \\ -1 & 0 & -2 & 0 & 0 \\ \hline 1 & -2 & 1 & -4 & 4 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & -3 & 3 \\ 0 & 0 & 0 & 1 & 0 \\ \hline 0 & 0 & 0 & 0 & 1 \end{array}\right]$$

$$\left[\begin{array}{cc|c} -2 & -4 & 4 \\ 1 & 0 & 0 \\ 0 & -3 & 3 \\ 0 & 1 & 0 \end{array}\right] \begin{cases} \xrightarrow[\substack{[(\frac{-1}{2})\mathbf{1}] \\ [(4)\mathbf{1}+\mathbf{2}] \\ [(-4)\mathbf{1}+\mathbf{3}]}]{\boldsymbol{\tau}} \left[\begin{array}{cc|c} 1 & 0 & 0 \\ \frac{-1}{2} & -2 & 2 \\ 0 & -3 & 3 \\ 0 & 1 & 0 \end{array}\right] \xrightarrow[\substack{[(1)\mathbf{2}+\mathbf{3}] \\ [(\frac{-1}{3})\mathbf{2}]}]{\boldsymbol{\tau}} \left[\begin{array}{cc|c} 1 & 0 & 0 \\ \frac{-1}{2} & \frac{2}{3} & 0 \\ 0 & 1 & 0 \\ 0 & \frac{-1}{3} & 1 \end{array}\right] \\ \xrightarrow[\substack{[(\frac{-1}{2})\mathbf{1}] \\ [(4)\mathbf{1}+\mathbf{2}] \\ [(-4)\mathbf{1}+\mathbf{3}]}]{\boldsymbol{\tau}} \left[\begin{array}{cc|c} 1 & 0 & 0 \\ \frac{-1}{2} & -2 & 2 \\ 0 & -3 & 3 \\ 0 & 1 & 0 \end{array}\right] \xrightarrow[\substack{[(\frac{-1}{2})\mathbf{2}] \\ [(\frac{1}{2})\mathbf{2}+\mathbf{1}] \\ [(-2)\mathbf{2}+\mathbf{3}]}]{\boldsymbol{\tau}} \left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \frac{3}{4} & \frac{3}{2} & 0 \\ \frac{-1}{4} & \frac{-1}{2} & 1 \end{array}\right] \end{cases}$$

Questions of the Lecture 12

(L-12) Question 1.

(a) Find a parametric representation of the line passing through the points $\boldsymbol{x}_P = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $\boldsymbol{x}_Q = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$.

(b) Find an implicit representation of the same line.

(L-12) Question 2.

(a) Find a parametric representation of the line passing through the points $\boldsymbol{x}_P = (1, \quad -3, \quad 1,)$ and $\boldsymbol{x}_Q = (-2, \quad 4, \quad 5,)$.

(b) Find an implicit representation (Cartesian equations) for the same line.

(L-12) Question 3.

(a) Parametric equation of a line parallel to $2x - 3y = 5$ that goes through $(1, 1)$.

(b) Find a implicit representation for the line.

(L-12) Question 4.

(a) Find the parametric equations of the plane that passes through the point $(0, \quad 1, \quad 1,)$ and is parallel to the vectors $(0, \quad 1, \quad 2,)$ and $(1, \quad 1, \quad 0,)$.

(b) Write the implicit equation of the same plane.

(L-12) Question 5.

(a) Find a parametric equation of the plane that passes through the point $(2, \quad 1, \quad 3,)$ and is perpendicular to the vector $(3, \quad 1, \quad 1,)$.
(b) Write the implicit equation of the same plane.

(L-12) Question 6. Consider the system of equations $\mathbf{A}\boldsymbol{x} = \boldsymbol{b}$, where

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 0 & 1 & 1 \\ 0 & 0 & 2 & 3 & 1 \\ 0 & 0 & 1 & 4 & 2 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}, \qquad \boldsymbol{b} = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix}.$$

(a) (1^{pts}) Find the solution to the system.
(b) (0.5^{pts}) Explain why the solution set to the previous system is a line in $\mathbb{R}^5$. Find a direction vector (a vector parallel to the line) and any point on that line.
(c) (1^{pts}) Find the set of vectors perpendicular to the previous direction vector. Prove that this set of perpendicular vectors forms a four-dimensional vector subspace. Find a basis for this subspace.

1 Outline of Lesson 13

Outline of Lesson 13

- Projections
- Projection Matrices

2 Direct Sum of Subspaces

$\mathbb{R}^n$ is a *direct sum* of $\mathcal{A}$ and $\mathcal{B}$ $\qquad (\mathbb{R}^n = \mathcal{A} \oplus \mathcal{B})$

if every $\boldsymbol{x} \in \mathbb{R}^n$ has a **unique** representation $\boldsymbol{x} = \boldsymbol{a} + \boldsymbol{b}$,

with $\boldsymbol{a} \in \mathcal{A}$ and $\boldsymbol{b} \in \mathcal{B}$.

Example $\boxed{\mathbb{R}^n = \mathcal{C}\left(\mathbf{A}^\intercal\right) \oplus \mathcal{N}\left(\mathbf{A}\right)}$

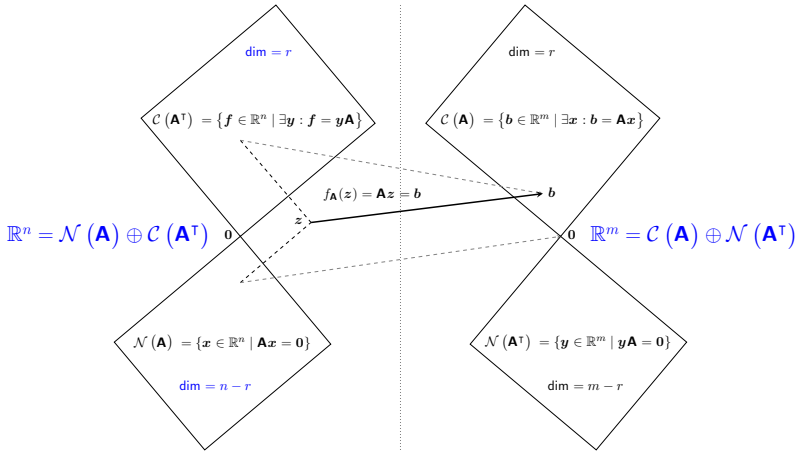
$$\begin{bmatrix} \mathbf{A} \\ \hline \mathbf{I} \end{bmatrix} = \begin{array}{c} \begin{bmatrix} 1 & 2 & 5 \\ 2 & 4 & 10 \end{bmatrix} \\ \hline \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{array} \rightarrow \begin{array}{c} \begin{bmatrix} 1 & 0 & 0 \\ 2 & 0 & 0 \end{bmatrix} \\ \hline \begin{bmatrix} 1 & -2 & -5 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{array} \Rightarrow \text{Basis of } \mathbb{R}^3; \left[\begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix}; \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}; \begin{pmatrix} -5 \\ 0 \\ 1 \end{pmatrix} \right]$$

$$\forall \boldsymbol{x} \in \mathbb{R}^3, \ \exists c_1, c_2, c_3 \ \Big| \ \boldsymbol{x} = c_1 \begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix} + c_2 \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} + c_3 \begin{pmatrix} -5 \\ 0 \\ 1 \end{pmatrix} = \boldsymbol{a} + \boldsymbol{b}$$

where $\boldsymbol{a} \in \mathcal{C}\left(\mathbf{A}^\intercal\right)$ and $\boldsymbol{b} \in \mathcal{N}\left(\mathbf{A}\right)$.

Also $\boxed{\mathbb{R}^m = \mathcal{C}\left(\mathbf{A}\right) \oplus \mathcal{N}\left(\mathbf{A}^\intercal\right)}$

3 The Big Picture: Direct Sum of Orthogonal Complements



$$\mathcal{C}\left(\mathbf{A}^\intercal\right) \perp \mathcal{N}\left(\mathbf{A}\right) \qquad \mathcal{C}\left(\mathbf{A}\right) \perp \mathcal{N}\left(\mathbf{A}^\intercal\right)$$

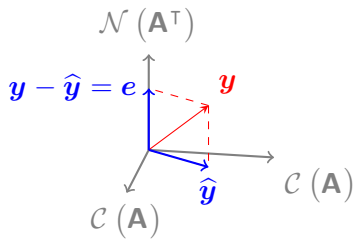
$$\boldsymbol{f} \cdot \boldsymbol{x} = \boldsymbol{y}\mathbf{A}\boldsymbol{x} = \boldsymbol{y} \cdot \boldsymbol{0} \qquad \boldsymbol{y} \cdot \boldsymbol{b} = \boldsymbol{y}\mathbf{A}\boldsymbol{x} = \boldsymbol{0} \cdot \boldsymbol{x}$$

4 Orthogonal Projection onto $\mathcal{C}(\mathbf{A})$

Consider $\underset{m\times n}{\mathbf{A}}$; since $\mathbb{R}^m = \mathcal{C}(\mathbf{A}) \oplus \mathcal{N}(\mathbf{A}^\intercal)$, for any $\boldsymbol{y} \in \mathbb{R}^m$

$$\boldsymbol{y} = \widehat{\boldsymbol{y}} + \boldsymbol{e}; \qquad (\boldsymbol{e} = \boldsymbol{y} - \widehat{\boldsymbol{y}})$$

where $\boxed{\widehat{\boldsymbol{y}} \in \mathcal{C}(\mathbf{A}) \text{ and } \boldsymbol{e} \perp \widehat{\boldsymbol{y}}}$, thus $\boldsymbol{e} \in \mathcal{N}(\mathbf{A}^\intercal)$.



How to compute $\widehat{\boldsymbol{y}} \in \mathcal{C}(\mathbf{A})$?

5 Normal Equations

Consider $\underset{m\times n}{\mathbf{A}}$. We want to find the decomposition $\boxed{\boldsymbol{y} = \widehat{\boldsymbol{y}} + \boldsymbol{e}}$ where

$$\widehat{\boldsymbol{y}} \in \mathcal{C}(\mathbf{A}) \qquad \text{and} \qquad (\widehat{\boldsymbol{y}} - \boldsymbol{y}) \in \mathcal{N}(\mathbf{A}^\intercal)$$

Thus

$$\mathbf{A}\widehat{\boldsymbol{x}} = \widehat{\boldsymbol{y}} \qquad \Leftrightarrow \qquad (\mathbf{A}\widehat{\boldsymbol{x}} - \boldsymbol{y}) \in \mathcal{N}(\mathbf{A}^\intercal)$$

That is

$$\mathbf{A}\widehat{\boldsymbol{x}} = \widehat{\boldsymbol{y}} \quad \Leftrightarrow \quad \mathbf{A}^\intercal(\mathbf{A}\widehat{\boldsymbol{x}} - \boldsymbol{y}) = \mathbf{0} \quad \Leftrightarrow \quad \boxed{(\mathbf{A}^\intercal\mathbf{A})\widehat{\boldsymbol{x}} = \mathbf{A}^\intercal\boldsymbol{y}}$$

Equivalent systems! $\Rightarrow \mathcal{N}(\mathbf{A}) = \mathcal{N}(\mathbf{A}^\intercal\mathbf{A}) \Rightarrow \mathrm{rg}(\mathbf{A}) = \mathrm{rg}(\mathbf{A}^\intercal\mathbf{A})$

$\boxed{\text{unique solution } \widehat{\boldsymbol{x}} \text{ if and only if } \mathbf{A} \text{ has independent columns}}$

6 Solution to the Normal Equations (Full Column Rank)

$\boxed{\mathbf{A}^\intercal\mathbf{A}\widehat{\boldsymbol{x}} = \mathbf{A}^\intercal\boldsymbol{y}}$ ($\mathbf{A}$ is full column rank)

The Solution $\quad \widehat{\boldsymbol{x}} = (\mathbf{A}^\intercal\mathbf{A})^{-1}\mathbf{A}^\intercal\boldsymbol{y}$

The Projection $\quad \widehat{\boldsymbol{y}} = \mathbf{A}\widehat{\boldsymbol{x}} = \mathbf{A}(\mathbf{A}^\intercal\mathbf{A})^{-1}\mathbf{A}^\intercal\boldsymbol{y}$

The Projection Matrix $\quad \mathbf{P} = \mathbf{A}(\mathbf{A}^\intercal\mathbf{A})^{-1}\mathbf{A}^\intercal$

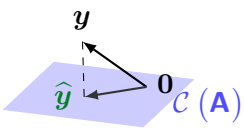
$$\widehat{\boldsymbol{y}} = \mathbf{P}\boldsymbol{y}$$

$\mathbf{P}$: Symmetric and Idempotent.

7 Projection Matrix

$$\boxed{\mathbf{P} = \mathbf{A}(\mathbf{A}^\intercal\mathbf{A})^{-1}\mathbf{A}^\intercal}$$

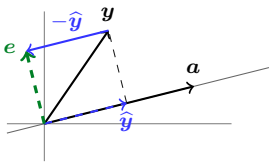
The Projection $\mathbf{P}\boldsymbol{y}$ *is the point* $\widehat{\boldsymbol{y}}$ *closest to* $\boldsymbol{y}$



Extreme Cases:

- If $\boldsymbol{y} \in \mathcal{C}(\mathbf{A})$ then $\mathbf{P}\boldsymbol{y} =$
- If $\boldsymbol{y} \perp \mathcal{C}(\mathbf{A})$ then $\mathbf{P}\boldsymbol{y} =$

8 Projection onto a Line



line$= \mathcal{C}(\mathbf{A})$; $\mathbf{A} = [a]$; $a \neq 0$

We want to find the point $\widehat{y}$ on the line that is closest to y

$\widehat{y} \in \mathcal{C}([a]) \quad \perp \quad e = (y - \widehat{y}) \in \mathcal{N}([a]^\mathsf{T})$.

$\widehat{y}$ is some multiple of a: $\quad \widehat{y} = [a](\widehat{x},)$

How: $\quad [a]^\mathsf{T}[a]\widehat{x} = [a]^\mathsf{T} y$

The Solution $\quad \widehat{x} = \quad ([a]^\mathsf{T}[a])^{-1}[a]^\mathsf{T} y$

The Projection $\quad \widehat{y} = [a]\widehat{x} = [a]([a]^\mathsf{T}[a])^{-1}[a]^\mathsf{T} y$

The Projection Matrix $\quad \mathbf{P} = [a]([a]^\mathsf{T}[a])^{-1}[a]^\mathsf{T}$

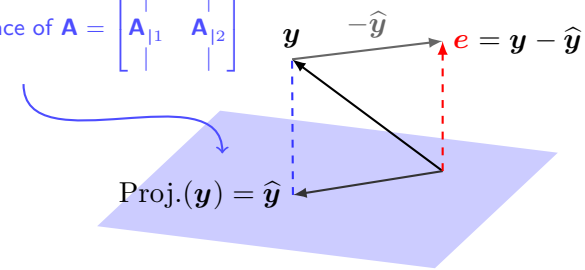
9 Projection onto a plane

Why project?

So we will solve

$$\mathbf{A}x = \Big(\text{Proj. of } y \text{ onto } \mathcal{C}(\mathbf{A})\Big).$$

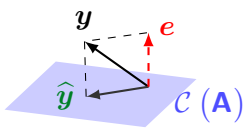
$\mathcal{C}(\mathbf{A})$: Column Space of $\mathbf{A} = \begin{bmatrix} | & | \\ \mathbf{A}_{|1} & \mathbf{A}_{|2} \\ | & | \end{bmatrix}$



$(y - \widehat{y}) = e \perp \mathcal{C}(\mathbf{A})$ $\quad$ ...that's the fundamental fact.

10 Normal Equations

What's the projection of y onto the column space of $\mathbf{A} = \begin{bmatrix} | & | \\ \mathbf{A}_{|1} & \mathbf{A}_{|2} \\ | & | \end{bmatrix}$?



$$\widehat{y} = (\mathbf{A}_{|1})\widehat{x_1} + (\mathbf{A}_{|2})\widehat{x_2} = \mathbf{A}\widehat{x}$$

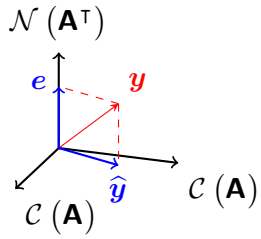
"Find a combination of the columns such that $e \perp \mathcal{C}(\mathbf{A})$"

$$e \perp \mathcal{C}(\mathbf{A}) \quad \Rightarrow \quad e \in$$

$$\mathbf{A}^\mathsf{T} e = \mathbf{A}^\mathsf{T}(y - \widehat{y}) \quad = \quad \mathbf{A}^\mathsf{T}(y - \mathbf{A}\widehat{x}) = 0 \quad \Leftrightarrow \quad \boxed{(\mathbf{A}^\mathsf{T}\mathbf{A})\widehat{x} = \mathbf{A}^\mathsf{T} y}$$

11 Two Projections

y has a component $\widehat{y}$ in $\mathcal{C}(\mathbf{A})$, and another component e in $\mathcal{C}(\mathbf{A})^\perp$.



$$\widehat{y} + e = y$$

$\widehat{y} = \mathbf{P}y$ $\quad$ is the projection onto $\mathcal{C}(\mathbf{A})$

$e = (\mathbf{I} - \mathbf{P})y$ $\quad$ projection onto $\mathcal{C}(\mathbf{A})^\perp$

Questions of the Lecture 13

(L-13) Question 1. Project the first vector orthogonally onto the line spanned by the second vector. Verify that $\boldsymbol{e}$ is perpendicular to $\boldsymbol{a}$. Find the projection matrix $\mathbf{P} = [\boldsymbol{a}]\left([\boldsymbol{a}]^\intercal[\boldsymbol{a}]\right)^{-1}[\boldsymbol{a}]^\intercal$ onto the line through each vector $\boldsymbol{a}$. Verify in each case that $\mathbf{P}^2 = \mathbf{P}$. Multiply $\mathbf{P}\boldsymbol{b}$ in each case to compute the projection $\widehat{\boldsymbol{b}}$.

(a) $\boldsymbol{b} = \begin{pmatrix}2\\1\end{pmatrix}$; $\boldsymbol{a} = \begin{pmatrix}3\\2\end{pmatrix}$.

(b) $\boldsymbol{b} = \begin{pmatrix}2\\1\end{pmatrix}$; $\boldsymbol{a} = \begin{pmatrix}3\\0\end{pmatrix}$.

(c) $\boldsymbol{b} = \begin{pmatrix}1\\1\\4\end{pmatrix}$; $\boldsymbol{a} = \begin{pmatrix}1\\2\\-1\end{pmatrix}$.

(d) $\boldsymbol{b} = \begin{pmatrix}1\\1\\4\end{pmatrix}$; $\boldsymbol{a} = \begin{pmatrix}3\\3\\12\end{pmatrix}$.

(Hefferon, 2008, exercise 1.6 from section VI.1.)

(L-13) Question 2. Project the vector orthogonally onto the line.

(a) $\begin{pmatrix}2\\-1\\4\end{pmatrix}$, The line : $\left\{\boldsymbol{v}\in\mathbb{R}^3 \;\middle|\; \exists \boldsymbol{p}\in\mathbb{R}^1,\ \boldsymbol{v} = \begin{bmatrix}-3\\1\\-3\end{bmatrix}\boldsymbol{p}\right\}$.

(b) $\begin{pmatrix}-1\\-1\end{pmatrix}$, the line described by the equation $y = 3x$.

(L-13) Question 3. Although diagrams have guided our development of the topic, we are not restricted to spaces that we can draw. In $\mathbb{R}^4$ project the vector onto the line.

$$\begin{pmatrix}1\\2\\1\\3\end{pmatrix};\quad \left\{\boldsymbol{v}\in\mathbb{R}^4 \;\middle|\; \exists \boldsymbol{p}\in\mathbb{R}^1,\ \boldsymbol{v} = \begin{bmatrix}-1\\1\\-1\\1\end{bmatrix}\boldsymbol{p}\right\}.$$

(L-13) Question 4.

(a) Project the vector $\boldsymbol{b} = (1,\quad 1,)$ onto the lines spanned by $\boldsymbol{a}_1 = (1,\quad 0,)$ and $\boldsymbol{a}_2 = (1,\quad 2,)$. Add the projections: $\widehat{\boldsymbol{b}_1} + \widehat{\boldsymbol{b}_2}$. The projections do not add to $\boldsymbol{b}$ because the vectors $\boldsymbol{a}_1$ and $\boldsymbol{a}_2$ are not orthogonal.

(b) The projection of $\boldsymbol{b}$ onto the plane spanned by $\boldsymbol{a}_1$ and $\boldsymbol{a}_2$ will equal $\boldsymbol{b}$. Find $\mathbf{P} = \mathbf{A}(\mathbf{A}^\intercal\mathbf{A})^{-1}\mathbf{A}^\intercal$ for $\mathbf{A} = [\boldsymbol{a}_1;\, \boldsymbol{a}_2;]$.

(Strang, 2003, exercise 8–9 from section 4.2.)

(L-13) Question 5.

(a) If $\mathbf{P}^2 = \mathbf{P}$ show that $(\mathbf{I}-\mathbf{P})^2 = \mathbf{I}-\mathbf{P}$. When $\mathbf{P}$ projects onto the column space of $\mathbf{A}$, $(\mathbf{I}-\mathbf{P})$ projects onto the _____.

(b) If $\mathbf{P}^\intercal = \mathbf{P}$ show that $(\mathbf{I}-\mathbf{P})^\intercal = \mathbf{I}-\mathbf{P}$.

(Strang, 2003, exercise 17 from section 4.2.)

(L-13) Question 6.

(a) Compute the projection matrices $\mathbf{P} = [\boldsymbol{a}]\left([\boldsymbol{a}]^\intercal[\boldsymbol{a}]\right)^{-1}[\boldsymbol{a}]^\intercal$ onto the lines through $\boldsymbol{a}_1 = (-1,\quad 2,\quad 2,)$ and $\boldsymbol{a}_2 = (2,\quad 2,\quad -1,)$. Show that $\boldsymbol{a}_1 \perp \boldsymbol{a}_2$. Multiply those projection matrices and explain why their product $\mathbf{P}_1\mathbf{P}_2$ is what it is.

(b) Project $\boldsymbol{b} = (1,\quad 0,\quad 0,)$ onto the lines through $\boldsymbol{a}_1$, and $\boldsymbol{a}_2$ and also onto $\boldsymbol{a}_3 = (2,\quad -1,\quad 2,)$. Add the three projections $\widehat{\boldsymbol{b}_1} + \widehat{\boldsymbol{b}_2} + \widehat{\boldsymbol{b}_3}$.

(c) Find the projection matrix $\mathbf{P}_3$ onto $\mathcal{L}([\boldsymbol{a}_3;]) = \mathcal{L}([(2,\quad -1,\quad 2,);])$. Verify that $\mathbf{P}_1 + \mathbf{P}_2 + \mathbf{P}_3 = \mathbf{I}$. The basis $\boldsymbol{a}_1$, $\boldsymbol{a}_2$, $\boldsymbol{a}_3$ is orthogonal!

(Strang, 2003, exercise 5–7 from section 4.2.)

(L-13) Question 7. Project $\boldsymbol{b}$ onto the column space of $\mathbf{A}$ by solving $\mathbf{A}^\intercal\mathbf{A}\widehat{\boldsymbol{x}} = \mathbf{A}^\intercal\boldsymbol{b}$ and then computing $\widehat{\boldsymbol{b}} = \mathbf{A}\widehat{\boldsymbol{x}}$. Find $\boldsymbol{e} = \boldsymbol{b} - \widehat{\boldsymbol{b}}$.

(a) $\mathbf{A}_1 = \begin{bmatrix}1&1\\0&1\\0&0\end{bmatrix}$ and $\boldsymbol{b}_1 = \begin{pmatrix}2\\3\\4\end{pmatrix}$

(b) $\mathbf{A}_2 = \begin{bmatrix}1&1\\1&1\\0&1\end{bmatrix}$ and $\boldsymbol{b}_2 = \begin{pmatrix}4\\4\\6\end{pmatrix}$

(c) Compute the projection matrices $\mathbf{P}_1$ and $\mathbf{P}_2$ onto the column spaces. Verify that $\mathbf{P}_1\boldsymbol{b}_1$ gives the first projection $\widehat{\boldsymbol{b}_1}$. Also verify $(\mathbf{P}_2)^2 = \mathbf{P}_2$.

(Strang, 2003, exercise 11–12 from section 4.2.)

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